UNIT – II : EARTH SLOPE STABILITY

1. Introduction

- A **slope** = an inclined ground surface – natural (hill, valley side) or man-made (embankment, cutting, earth dam).
- Soil mass on a slope tends to slide down under its own weight (gravity). Slope stability analysis checks whether the shear strength available along a possible slip surface is enough to resist this.
- Aim of analysis: find the Factor of Safety (F) and the most dangerous (critical) slip surface (one with minimum F).

F = Shear strength available (τ_f) / Shear stress mobilised (τ_m)

- $F < 1 \rightarrow$ failure; $F = 1 \rightarrow$ limiting equilibrium; **F = 1.25 – 1.5** is generally adopted for permanent slopes (1.3 – 1.5 for earth dams; 1.0 – 1.2 temporary).

2. Classification of slopes

- **Infinite slope** – constant inclination β, very large extent, ends ignored; failure plane shallow & parallel to surface (long hill side).
- **Finite slope** – limited height H with crest & toe; slip surface curved/plane (embankments, dams, cuttings).

3. Causes of slope failure

(a) Increase in shear stress

- Extra load on crest; steepening of slope
- Removal of toe support (excavation, river erosion)
- Rain saturation; earthquake, blasting

(b) Decrease in shear strength

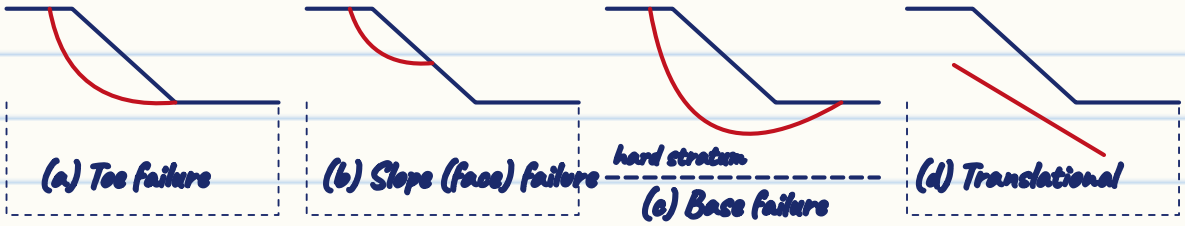
- Rise in pore water pressure u ($\sigma' = \sigma - u$ falls)
- Softening/swelling of clay, loss of c
- Weathering; progressive failure; sudden drawdown

Rule: any process that raises τ or lowers s $\Rightarrow$ F decreases $\Rightarrow$ slide!

4. Assumptions in stability analysis

- Problem treated as **plane strain** (2-D) – unit length along the slope.
- Homogeneous, isotropic soil; Mohr–Coulomb $s = c + \sigma' \tan\varphi$; soil above slip surface moves as a rigid body.

5. Types of failures of finite slopes



(a),(b),(c) = rotational slip (circular arc) – homogeneous cohesive soils
(d) = plane/wedge slip – along a weak layer, or c-φ=0 granular soil

Fig. 1 – Types of slope failure (red = slip surface)

(i) Rotational failure (circular arc)

- **Toe failure** – arc passes through the toe. Common in steep slopes ($\beta > 53°$) and soils with $\varphi > 0$.
- **Slope / face failure** – arc cuts the slope above toe; occurs when a hard layer exists at the toe level or for flat slopes with firm soil near the toe.
- **Base (deep) failure** – arc passes below the toe into the foundation; occurs in soft clays ($\varphi = 0$) with flatter slopes ($\beta < 53°$) with no hard stratum nearby.

(ii) Translational failure

- Slip surface is almost **plane**, parallel to the slope, at a shallow depth – infinite slopes, or when a weak stratum (soft clay seam, rock surface) is present.

(iii) Compound / wedge failure

- Curved + plane portions when a hard stratum lies at some depth; wedge sliding along 2–3 planes (rock slopes, weak seams).

Depth factor $n_d = D/H$; depth D of firm stratum below toe (H = height of slope). Base failure when D is large.

6. Methods of analysis

- Infinite slope: element equilibrium. Finite planar: Culmann. Finite circular: Swedish circle ($\varphi=0$), slices (Fellenius/Bishop), friction circle, Taylor charts.

INFINITE SLOPES

7. Infinite slope in cohesionless soil (dry)

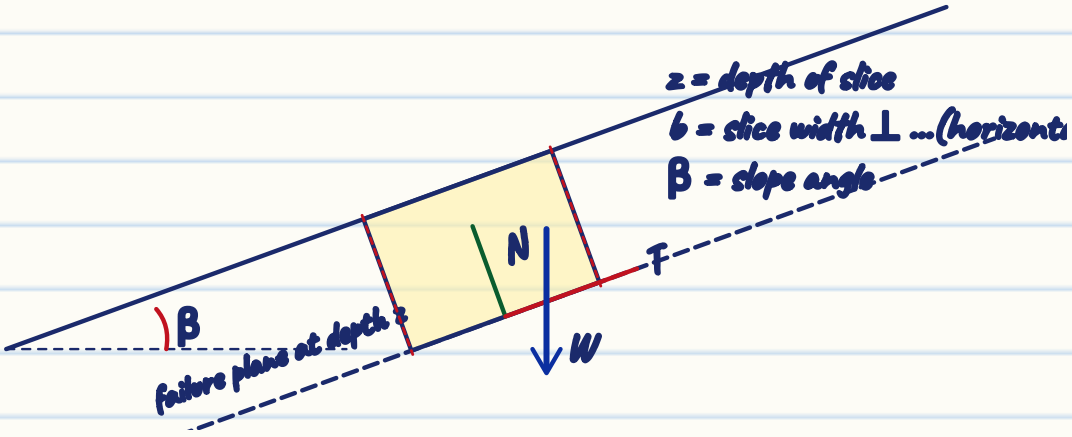


Fig. 2 – Forces on a slice of an infinite slope

- Consider a slice of width b (horizontal) and depth z. Side forces on the two vertical faces are equal & opposite (same depth, same conditions) – they cancel.
- Weight of slice: $W = \gamma\, b\, z$ (per unit length).
- Length of failure plane: $l = b / \cos\beta$.

$$N = W\cos\beta = \gamma\, b\, z\cos\beta \;;\; T = W\sin\beta = \gamma\, b\, z\sin\beta$$

Normal stress $\sigma = N/l = \gamma\, z\cos^2\beta$; Shear stress $\tau = T/l = \gamma\, z\sin\beta\cos\beta$

For cohesionless dry soil: $s = \sigma\tan\varphi = \gamma\, z\cos^2\beta\tan\varphi$

$$F = s/\tau = (\gamma\, z\cos^2\beta\tan\varphi)/(\gamma\, z\sin\beta\cos\beta) = \tan\varphi / \tan\beta$$

- F is **independent of depth** z and of unit weight γ.
- Limiting stable slope: $F = 1 \Rightarrow \beta = \varphi$ (angle of repose). A dry sand slope can never be steeper than φ.
- For $F = 1.5$ and $\varphi = 30°$ $\Rightarrow \tan\beta = 0.5774/1.5 \Rightarrow \beta = 21°$.

Same answer if the slope is completely submerged in still water (no seepage) – because both γ' and u act, ratio unchanged.

8. Infinite slope in c–φ soil (no seepage)

- Shear strength: $s = c + \sigma\tan\varphi = c + \gamma\, z\cos^2\beta\tan\varphi$
- Mobilised shear stress: $\tau = \gamma\, z\sin\beta\cos\beta$

$$F = c/(\gamma\, z\sin\beta\cos\beta) + \tan\varphi/\tan\beta$$

$$F = F_c + F_\varphi \;,\; F_c = c/(\gamma\, z\cos^2\beta\tan\beta)$$

- F is made of a **cohesion part** (decreases with depth z) and a **friction part** (independent of z).
- F reduces as z increases ⇒ deeper failure planes are more critical for c–φ soil until a rock/firm layer.

Critical depth (Hc) – depth at which F = 1

$$1 = c/(\gamma H_c \cos^2\beta (\tan\beta)) + \tan\varphi/\tan\beta$$

$$H_c = c / [\gamma \cos^2\beta (\tan\beta - \tan\varphi)]$$

Stability number (S_n) for infinite slope

$$S_n = c / (\gamma H_c) = \cos^2\beta (\tan\beta - \tan\varphi)$$

- For cohesive soil φ = 0: $H_c = c/(\gamma \sin\beta \cos\beta) = 2c / (\gamma \sin 2\beta)$.
- Hence for given F, the safe depth: $H = c / [\gamma \cos^2\beta (\tan\beta - \tan\varphi_m)]$ where $c_m = c/F_c$, $\tan\varphi_m = \tan\varphi/F_\varphi$ are mobilised values ($F_c = F_\varphi = F$).

9. Infinite slope with seepage parallel to the slope

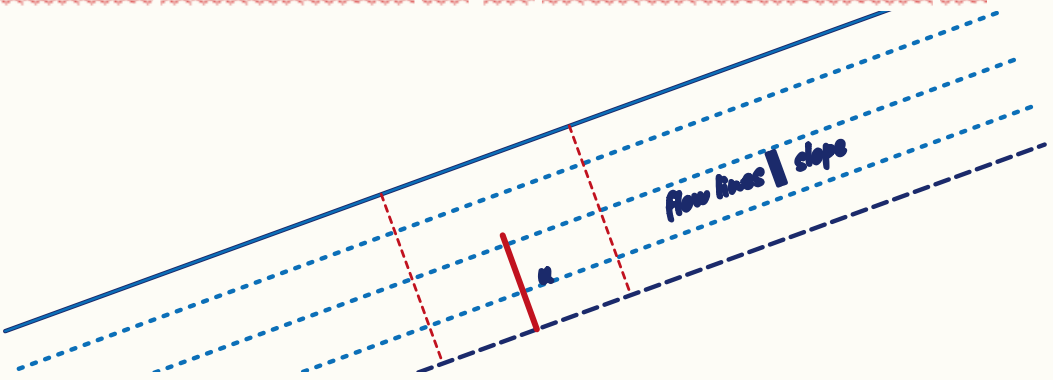


Fig. 3 – Water table at ground surface, flow parallel to slope

- Flow lines ∥ slope, equipotentials ⊥ slope. Pore pressure on the failure plane = water head along equipotential: $u = \gamma_w z \cos^2\beta$.
- Saturated unit weight γ_{sat} used for W: $\sigma = \gamma_{sat} z \cos^2\beta$; $\sigma' = \sigma - u = (\gamma_{sat} - \gamma_w) z \cos^2\beta = \gamma' z \cos^2\beta$.

$$F = c' / (\gamma_{sat} z \sin\beta \cos\beta) + (\gamma'/\gamma_{sat}) \cdot (\tan\varphi'/\tan\beta)$$

- Since $\gamma' \approx \gamma_{sat}/2$, the friction term is nearly halved – seepage parallel to slope reduces safety by about 50 % for cohesionless soil.
- For c' = 0: $F = (\gamma'/\gamma_{sat}) \tan\varphi'/\tan\beta$ ⇒ stable slope angle is only about **half of φ'**: $\tan\beta \approx ½ \tan\varphi'$.

- If the water table is at depth m z (m = fraction of z that is saturated): $u = \gamma_w (1-m)$... $F = c'/(\gamma z \sin\beta \cos\beta) + [1 - r_u \sec^2\beta] \tan\varphi'/\tan\beta$ where $r_u = u/\gamma z$ (pore pressure ratio).

Condition (c = 0)	F
Dry slope	$\tan\varphi/\tan\beta$
Fully submerged, no seepage	$\tan\varphi/\tan\beta$
Seepage ∥ slope, WT at surface	$(\gamma'/\gamma_{sat}) \tan\varphi/\tan\beta \approx \frac{1}{2} \tan\varphi/\tan\beta$

Important: dry/submerged → $\beta \leq \varphi$. With seepage → $\beta \leq \sim \varphi/2$! This is why slopes fail after heavy rain.

Example 1

An infinite slope of inclination 20° is made of soil with $c' = 10$ kN/m², $\varphi' = 30°$, $\gamma_{sat} = 19$ kN/m³. Find F against a failure plane at depth 3 m parallel to slope if there is seepage parallel to slope with water table at ground surface. ($\gamma_w = 9.81$)

Solution: $\gamma' = 19 - 9.81 = 9.19$ kN/m³; $\sin20° = 0.342$, $\cos20° = 0.940$, $\tan20° = 0.364$, $\tan30° = 0.577$

$$F = 10/(19\times3\times0.940\times0.342) + (9.19/19)(0.577/0.364)$$
$$= 0.546 + 0.484 \times 1.586 = 0.546 + 0.767$$

F = 1.31 (safe, > 1.25 ✓)

Check without seepage: $F = 0.546 + 1.586 \times (19/19) = 2.13$ (using $\gamma_{sat} = \gamma$) – the seepage almost halves F (2.13 → 1.31).

Exam tip: write the formula, substitute, and conclude "slope is safe / unsafe" as per $F \geq 1.25$.

FINITE SLOPES – CIRCULAR SLIP

10. Swedish Circle Method ($\varphi_u = 0$ analysis)

- Developed by Swedish geotechnical commission (Petterson, Fellenius) after the Gota-river quay failures (1916).
- Suitable for **saturated clay under undrained (short-term) condition:** $\varphi_u = 0$, shear strength = c_u (constant).

- Slip surface = circular arc; whole soil mass above it taken as one block rotating about centre O.

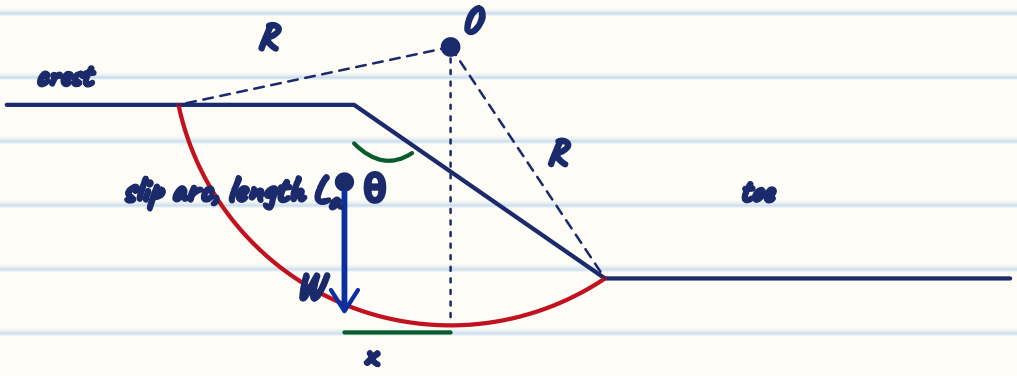


Fig. 4 – Swedish circle ($\varphi = 0$)

Derivation

- W = weight of the sliding mass (area × γ, per unit length), acts at its centroid, lever arm x from O → **Disturbing moment** $M_D = W \cdot x$
- Resisting force on arc = $c_u \times L_a$ where L_a = arc length = $R\theta$ (θ in radians) acts at distance R → **Resisting moment** $M_R = c_u L_a R = c_u R^2\theta$

$$F = M_R/M_D = c_u L_a R / (W x) = c_u R^2\theta / (W x)$$

- Trial circles are repeated with different centres; the smallest F = critical circle (F_{min}).

Tension cracks

- In cohesive soil a vertical **tension crack** forms at the crest to depth $z_c = 2c_u/\gamma$ (tensile strength = 0).
- Effect: (i) shear resistance along crack length is lost → arc length reduces to L_a' (arc only up to crack) and W reduces; (ii) if crack fills with water, a hydrostatic thrust $P_w = \frac{1}{2}\gamma_w z_c^2$ acts horizontally at $z_c/3$ from the crack bottom, adding to disturbing moment.

$$F = c_u L_a' R / (W x + P_w y_w)$$

- y_w = vertical distance from O to the line of action of P_w. Cracks reduce F → hence the critical circle must be tried with cracks.

Limitations of Swedish circle method

- Applicable only for $\varphi = 0$ soils (undrained clays).

- Does not consider that c_u may vary with depth, so a layered soil must be dealt with by slices.
- Trial-and-error: many circles required.

For c–φ soil, use method of slices (next pages).

Example 2 (Swedish circle)

A trial slip circle of radius 10 m is selected for an embankment on saturated clay (c_u = 30 kN/m², φ_u = 0). The area of the sliding mass is 35 m², γ = 20 kN/m³; centroid of the sliding mass is at horizontal distance 4.5 m from the centre O. The central angle of the arc is 103°. Find F.

Solution

- $W = A \times \gamma = 35 \times 20 =$ **700 kN/m**
- $\theta = 103° = 103 \times \pi/180 = 1.798$ rad ; $L_a = R\theta = 10 \times 1.798 =$ **17.98 m** ≈ 18 m
- $M_R = c_u L_a R = 30 \times 17.98 \times 10 = 5394$ kN·m/m
- $M_D = W x = 700 \times 4.5 = 3150$ kN·m/m

$$F = 5394 / 3150 = 1.71 > 1.5 \text{ (safe)}$$

If a water-filled tension crack of $z_c = 2\times30/20 = 3$ m existed: $P_w = ½ \times 9.81 \times 3^2 = 44$ kN/m (arm ≈ 8 m) → M_D increases by 352 → $F \approx 1.54$ (reduced).

II. Method of slices – Standard (Fellenius / Swedish) method

- For c–φ soils, non-homogeneous slopes, with seepage. Sliding mass above arc divided into **n vertical slices** of equal width b (≈ R/10).
- Each slice treated separately: forces on it –

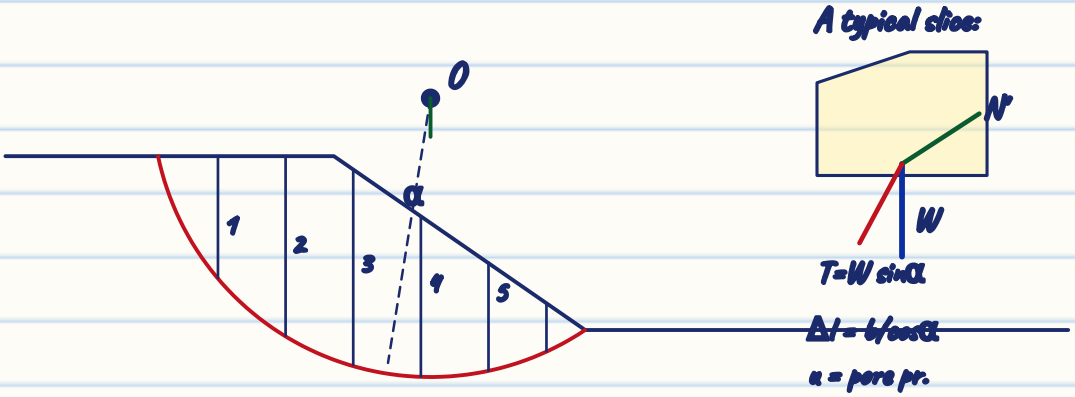


Fig. 5 – Slices (b = width, α = angle of base with horizontal)

- W (weight) ; $N = W\cos\alpha$ (normal) ; $T = W\sin\alpha$ (tangential). Inter-slice forces neglected (Fellenius assumption: resultant of side forces is parallel to the slice base) – so no change to the above.
- Slice base length $\Delta l = b / \cos\alpha$. α is +ve when base slopes up in same direction as the slope surface (toward toe), –ve on the other side of the vertical through O.

Derivation (effective stress)

- Shear resistance on slice base: $S = [\, c'\Delta l + (N - u\Delta l)\tan\varphi' \,] / F$
- Moments about O: driving = $\Sigma\, W R \sin\alpha$; resisting = $R \Sigma\, S$"

$$F = \Sigma\, [\, c'\Delta l + (W\cos\alpha - u\Delta l)\tan\varphi' \,] / \Sigma\, W\sin\alpha$$

- Total stress / undrained case ($\varphi = 0$, u not considered): $F = \Sigma\, c_u \Delta l / \Sigma\, W\sin\alpha = c_u L_a / \Sigma\, W\sin\alpha$. (Equivalent to Swedish circle)
- Pore pressure ratio: $r_u = u/(\gamma h) = u\, b / W \Rightarrow u\Delta l = r_u W / \cos\alpha$ … so $F = \Sigma[c'\Delta l + W(\cos\alpha - r_u \sec\alpha)\tan\varphi\,]/\Sigma\, W\sin\alpha$

Fellenius: simple, conservative (F is 5–20 % lower than Bishop), but can be too conservative for large pore pressures/flat arcs.

12. Bishop's simplified method (mention)

- Considers inter-slice normal forces E; assumes vertical shear forces between slices are zero ($X_n - X_{n+1} = 0$). Vertical equilibrium of each slice gives N' so that F appears on both sides – **iterative**.

$$F = \Sigma\, \{\, [\, c'b + (W - ub)\tan\varphi\,] / m_\alpha \,\} / \Sigma\, W\sin\alpha$$

$$m_\alpha = \cos\alpha\,(1 + \tan\alpha\tan\varphi'/F)$$

- Assume F (take Fellenius value) → calculate m_α → new F → repeat until same (2–3 trials).

Feature	Fellenius (ordinary)	Bishop simplified
Inter-slice forces	Ignored	Normal force E included, shear X = 0
Equations satisfied	Moment only	Moment + vertical force
Solution / result	Direct, conservative	Iterative, more accurate

Steps for method of slices

1. Draw the slope to scale; select a trial centre O and radius R; draw circle.
2. Divide into 6–10 slices (equal widths; a slice must have a single base soil).
3. Find $W = \gamma \times$ area, α, Δl, u (flow net) for each; tabulate $W\cos\alpha$, $W\sin\alpha$, $c'\Delta l$.
4. Compute F; repeat for other circles; plot contours of F – minimum F = critical.

Example 3 (Fellenius method)

For a trial circle, the sliding mass is divided into 5 slices of width 2 m. $c' = 10\ kN/m^2$, $\varphi' = 20°$, no pore pressure ($u = 0$). Find F.

Slice	W (kN/m)	α	$\Delta l = 2/\cos\alpha$	$W\cos\alpha$	$W\sin\alpha$
1	40	−8°	2.02	39.6	−5.6
2	90	6°	2.01	89.5	9.4
3	120	20°	2.13	112.8	41.0
4	95	35°	2.44	77.8	54.5
5	40	50°	3.11	25.7	30.6
Σ			**11.71**	**345.4**	**130.0**

$c\Sigma\Delta l = 10 \times 11.71 = 117.1$; $\Sigma W\cos\alpha\tan\varphi' = 345.4 \times 0.364 = 125.7$

$$F = (117.1 + 125.7)/130.0 = 1.87$$

FRICTION CIRCLE METHOD

13. Friction circle method (Taylor)

- A **graphical**/force-polygon method for c–φ soil (homogeneous), treats the slip mass as a whole (no slices). Developed by Taylor (1937).
- Three forces act on the sliding mass at failure: (1) **W** – weight through centroid, (2) **C** – resultant of mobilised cohesion on the arc, (3) **P** – resultant of normal & friction forces on arc.

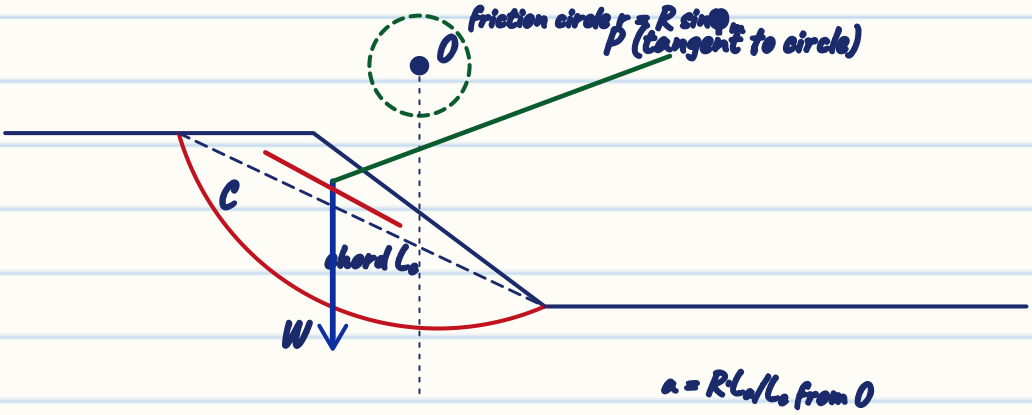


Fig. 6 – Friction circle method

Concept

- Mobilised friction angle $\varphi_m = \tan^{-1}(\tan\varphi / F_\varphi)$; mobilised cohesion $c_m = c / F_c$.
- The resultant P of normal & frictional forces at any point of the arc makes angle φ_m with the normal ⇒ it is tangent to a circle of radius $r_\varphi = R \sin\varphi_m$ concentric with slip circle – the **friction circle**. (Line of P assumed tangent to this circle.)
- Resultant cohesion force: $C = c_m \times L_c$ (L_c = chord length), parallel to chord, at perpendicular distance from O:

$$a = R \cdot L_a / L_c \ (L_a = \text{arc length})$$

Procedure (graphical)

1. Draw slope to scale. Choose trial circle (centre O, radius R); find area & centroid ⇒ $W = \gamma A$.
2. Assume a trial factor of safety against friction F_φ (say 1.5); calculate $\varphi_m = \tan^{-1}(\tan\varphi / F_\varphi)$ and draw friction circle of radius $R \sin\varphi_m$.
3. Draw chord AB, measure L_c; calculate arc length L_a; find $a = R\, L_a / L_c$. Draw line of action of cohesion C parallel to AB at distance a from O.
4. Extend W and C to intersect at a point; draw the line of P through this point tangent to friction circle (so that the moment of P about O has the same sense as W's, i.e. P resists).
5. Draw force triangle: W known in magnitude and direction; draw C ∥ chord and P along tangent line → closed triangle gives the value of C_m (mobilised cohesion force).
6. $c_m = C_m / L_c$; $F_c = c / c_m$.

7. If $F_c \neq F_\varphi$ assumed, repeat with new F_φ until $F_c = F_\varphi = F$ (plot F_c vs F_φ and read intersection with 45° line).

Special case $\varphi = 0$: friction circle vanishes ($r = 0$) → P passes through O → reduces to Swedish circle: $F = c\, L_a\, R/(W\, x)$.

Merits & limitations

Merits

- Whole mass treated – no slices
- Handles c–φ soil, with seepage by using effective stress & total weight
- Basis for Taylor's charts

Limitations

- Assumes homogeneous soil
- Trial & error and tedious graphics
- Tangent assumption of P is approximate (slightly conservative)

14. Taylor's stability number method

- Taylor (1948) analysed homogeneous slopes by friction-circle method and presented results as **dimensionless charts**, so no tedious trial circles are needed.
- Define the stability number :

$$S_n = c_m / (\gamma H) = c / (F_c \gamma H)$$

- c_m = mobilised (required) cohesion; H = height of slope; γ = unit weight. S_n depends on slope angle β, mobilised friction angle φ_m and depth factor $n_d = D/H$.
- Reciprocal $1/S_n = \gamma H / c_m$ is called **stability factor** (m). Critical height ($F = 1$): $H_c = c/(\gamma S_n)$.

Taylor's chart for $\varphi = 0$ (approx. values of S_n)

Slope β	90°	75°	60°	53°	45°	30°	15°
S_n	0.261	0.219	0.191	0.181	0.170	0.156	0.145

- $\beta > 53°$: always toe failure; S_n independent of depth factor.
- $\beta < 53°$: failure may be toe, slope or base type depending on depth factor $n_d = D/H$ (chart gives curves for n_d = 1, 1.25, 1.5, 2 … ∞). Maximum S_n for $\varphi = 0$ is 0.181 at $\beta = 53°$ ($n_d = \infty$ → midpoint circle).
- For $\varphi > 0$ (φ = 5°, 10°, 15°, 20°, 25°) curves of S_n vs β are given (toe circles) – S_n decreases as φ increases.

Procedure for c–φ soil (stability analysis)

1. Assume F_φ ; $\varphi_m = \tan^{-1}(\tan\varphi / F_\varphi)$.

2. From chart with β, φ_m (and n_d) read S_n.

3. $c_m = S_n \gamma H \Rightarrow F_c = c/c_m$.

4. Repeat until $F_c = F_\varphi = F$.

Design problem: given F, find H or β: $c_m = c/F$; $\varphi_m = \tan^{-1}(\tan\varphi/F) \Rightarrow S_n$ from chart $\Rightarrow H = c_m/(\gamma S_n) = c/(F\gamma S_n)$.

If only F against cohesion is wanted ($\varphi = 0$): $F = c / (S_n \gamma H)$ – straight, no trial!

Example 4 (Taylor)

A 10 m high cut slope of 1 H : 1 V ($\beta = 45°$) is in saturated clay with $c_u = 40$ kN/m², $\varphi_u = 0$, $\gamma = 19$ kN/m³. A hard stratum is at great depth ($n_d = \infty$). Find F and critical height.

Solution: Taylor's chart $\varphi = 0$, $\beta = 45° \Rightarrow S_n = 0.170$.

$$c_m = S_n \gamma H = 0.170 \times 19 \times 10 = 32.3 \text{ kN/m}^2$$

$$F = c/c_m = 40/32.3 = 1.24$$

$$H_c = c/(\gamma S_n) = 40/(19 \times 0.170) = 12.4 \text{ m}$$

EARTH DAMS – STABILITY OF SLOPES

15. Stability of slopes of earth dams

- Earth dam = embankment of compacted soil to store water; slopes must be stable **at all stages of life:** construction, reservoir full, drawdown.
- Shear strength parameters and pore pressures differ at each stage ⇒ four critical conditions are checked, using method of slices (circular arcs) – trial circles till F_{min}.

Condition	Slope	Analysis	Min F
(a) End of construction	Upstream & downstream	Total stress (UU) or effective with construction u	1.25
(b) Steady seepage (reservoir full)	**Downstream**	Effective stress (c', φ'), u from flow net	1.5
(c) Sudden drawdown	**Upstream**	Effective stress, u of full pool retained	1.2–1.3
(d) Partial pool	Upstream	Effective stress	1.5

(b) Steady seepage condition

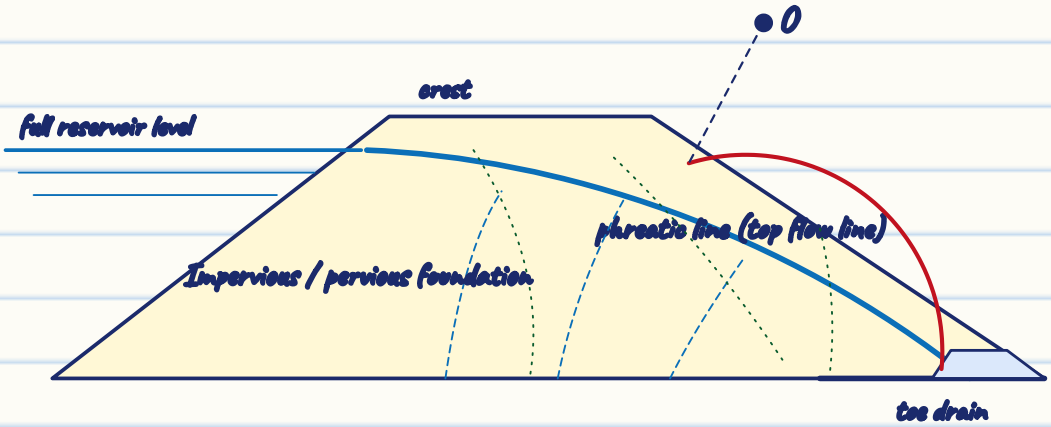


Fig. 7 – Earth dam, steady seepage : phreatic line, flow net & critical slip circle on d/s side

- Phreatic line (top flow line) = **Casagrande's parabola** corrected at entry; flow net drawn; pore pressure at base mid-point of each slice: $u = \gamma_w h_w$ (h_w = height from point up to the equipotential line meeting the point, measured to the phreatic level).
- Soil **above phreatic line:** moist γ ; **below:** saturated γ_{sat} (use in W). Shear strength in terms of **effective stress** (c', φ') from CD/CU tests:

$$F = \Sigma \left[c'\Delta l + (W\cos\alpha - u\Delta l)\tan\varphi' \right] / \Sigma\, W\sin\alpha$$

- Downstream slope is the critical one – seepage force acts outward; pore pressures are high near the toe ⇒ provide **toe drain / rock toe** and a filter to lower the phreatic line ⇒ F increases (typically from 1.1 to > 1.5).
- Alternatively use r_u (average pore pressure ratio = 0.3–0.5): $u\Delta l = r_u W \sec\alpha$.

(a) End of construction condition

- Pore pressures generated by compaction/loading have not yet dissipated (especially in clayey cores) – **undrained.**
- Analyse in **total stress** with UU test parameters (c_u, φ_u) – weights with moist γ: $F = \Sigma[c_u\Delta l + W\cos\alpha\tan\varphi_u]/\Sigma\, W\sin\alpha$. For saturated clay $\varphi_u = 0$ (Swedish circle).
- Or use effective-stress parameters with u from construction pore pressure (r_u from piezometers). Both upstream and downstream slopes checked; critical for dams on soft foundation clays.
- Reservoir empty ⇒ no water force on u/s face; stability improves later with consolidation (F ↑ with time).

(c) Sudden (rapid) drawdown condition

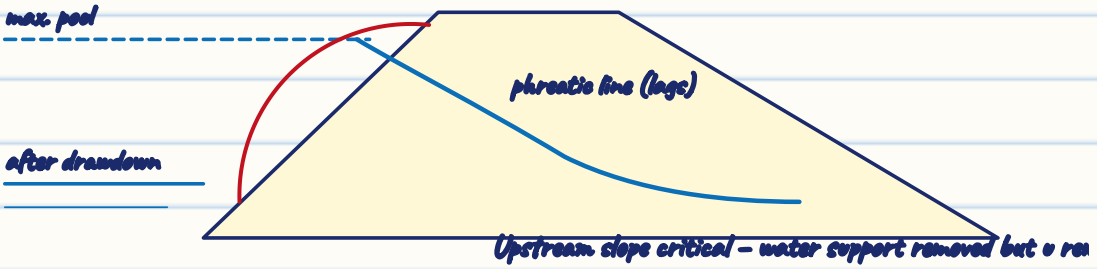


Fig. 8 – Sudden drawdown

- Reservoir level falls quickly (faster than soil can drain – in low-permeability shell/core). The water pressure on the u/s face disappears but pore pressures inside remain at **full-reservoir level** (no dissipation).
- Soil saturated ⇒ W uses γ_{sat} for all submerged soil (no buoyancy). Normal stress ($\sigma - u$) reduced and driving force increased ⇒ most critical for the upstream slope.
- Pore pressure can be obtained from the steady seepage flow net (before drawdown) – effective-stress method: $F = \Sigma[c'\Delta l + (W\cos\alpha - u\Delta l)\tan\varphi']/\Sigma W\sin\alpha$ with W from γ_{sat}; OR use the CU (consolidated-undrained) parameters in total stress analysis: $F = \Sigma[c_{cu}\Delta l + W\cos\alpha\tan\varphi_{cu}]/\Sigma W\sin\alpha$ (Bishop's / U.S. Army Corps two-stage method).
- Morgenstern (1963) charts give F directly for drawdown ratio ($\Delta H/H$); F is lowest when drawdown is quick and the soil is of low permeability (clay).
- Free-draining sand / gravel shells dissipate u quickly ⇒ drawdown is not critical for these.

Remedies: flatter u/s slope, rip-rap, pervious shell, controlled drawdown rate (≈ 15 cm/day).

Summary of key formulae

Case	Factor of safety
Infinite slope, $c = 0$	$\tan\varphi / \tan\beta$
Infinite slope, c–φ	$c/(\gamma z \sin\beta \cos\beta) + \tan\varphi/\tan\beta$
Infinite, seepage ∥ slope	$c'/(\gamma_{sat} z \sin\beta \cos\beta) + (\gamma'/\gamma_{sat}) \tan\varphi'/\tan\beta$
Swedish circle ($\varphi=0$)	$c_u L_a R/(W x)$
Fellenius slices	$\Sigma[c'\Delta l+(W\cos\alpha-u\Delta l)\tan\varphi']/\Sigma W \sin\alpha$
Bishop	$\Sigma\{[c'b+(W-ub)\tan\varphi']/m_\alpha\}/\Sigma W \sin\alpha$
Taylor	$F_c = c/(S_n \gamma H)$; $H_c = c/(\gamma S_n)$

Important questions (exam)

1. Explain the different types of slope failures with neat sketches. What are the causes of failure of slopes?
2. Derive the expression for factor of safety of an infinite slope in c–φ soil. How is it modified for seepage parallel to the slope?
3. Explain Swedish circle method for analysing stability of a slope in purely cohesive soil.
4. Describe the method of slices (Fellenius) for slope stability. Compare with Bishop's method.
5. Explain the friction circle method with neat sketch. Define Taylor's stability number and explain its use.
6. Discuss the stability of u/s and d/s slopes of an earth dam under (i) steady seepage (ii) sudden drawdown (iii) end of construction.
7. Numericals: infinite slope with seepage; Swedish circle with tension crack; Fellenius table; Taylor's H_c and F.